

## ARJUN GIRIRAJ DAY 07

1. What is a network address?

The network address identifier of a host or device on network.

2. What is a broadcast address?

Broadcast address is a network address used to transmit data to all devices in same network address.

3. What is a usable host address?

Usable host address are the IP addresses within the IPv4 subnet addresses.

4. What is the first usable IP address?

The First usable IP address is the first IP address which is in IPv4 subnet address ranges.

5. What is the last usable IP address?

The second last IP address which is in the last of the IPv4 subnet address ranges.

6. What does /24 mean?

/24 means 24 bits represent the network prefix.

7. What subnet mask corresponds to /24?

The subnet masks 255.255.255.0 corresponds to /24.

8. What is a default gateway?

The router sends data to when the data has to go outside the local network.

9. Why does a router interface need an IP address?

Router interface needs an IP address as it operates in the Layer 3 on network model.

10. What is a static IP configuration?

Static IP addressing is manually configured and kept consistent, it is used for servers, printers, etc.

11. What does "show ip interface brief" display?

It displays Interface, IP address, Status, Protocol.

12. What does "no shutdown" do conceptually on a Cisco interface?

The "no shutdown" command enables the interface.

13. What does interface status "up/up" generally indicate?

Shows the network interface fully operational.

14. Why should network configuration always be verified?

Security professionals always verify the configuration because security professionals have to work with IP address, subnet, router, packet-capture, etc.

15. Why is IPv4 configuration important in cybersecurity?

Security professionals work with IP addresses, subnets, routers, packet captures, Nmap results, Wireshark captures, incident response and threat hunting.

For 192.168.100.0/24 identify:

- Network Address: 192.168.100.0
- First Usable IP: 192.168.100.1
- Last Usable IP: 192.168.100.254
- Broadcast Address: 192.168.100.255

From YOUR OWN PC (ipconfig /all) record:

IPv4 Address -- 10.230.76.85

Subnet Mask -- 255.255.255.0

Default Gateway -- 10.230.76.191

DNS Server - 10.230.76.191

Attach screenshots of:

ipconfig /all

```
venom@VENOM:~$ ifconfig -a
lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    inet6 ::1 prefixlen 128 scopeid 0x10<host>
    loop txqueuelen 1000 (Local Loopback)
    RX packets 8939 bytes 835112 (815.5 KiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 8939 bytes 835112 (815.5 KiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

wlan0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 10.230.76.85 netmask 255.255.255.0 broadcast 10.230.76.255
    inet6 fe80::83a4:13e:661d:1ec prefixlen 64 scopeid 0x20<link>
    inet6 2401:4900:bfb7:ceb5:eca9:5603:9c4b:c5e5 prefixlen 64 scopeid 0x0
    <global>
    ether 10:6f:d9:7a:da:a9 txqueuelen 1000 (Ethernet)
    RX packets 70118 bytes 51065356 (48.6 MiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 41442 bytes 14999436 (14.3 MiB)
    TX errors 0 dropped 11 overruns 0 carrier 0 collisions 0
```

ping 127.0.0.1 ·

```
venom@VENOM:~$ ping 127.0.0.1
PING 127.0.0.1 (127.0.0.1) 56(84) bytes of data:
64 bytes from 127.0.0.1: icmp_seq=1 ttl=64 time=0.043 ms
64 bytes from 127.0.0.1: icmp_seq=2 ttl=64 time=0.038 ms
64 bytes from 127.0.0.1: icmp_seq=3 ttl=64 time=0.080 ms
64 bytes from 127.0.0.1: icmp_seq=4 ttl=64 time=0.121 ms
64 bytes from 127.0.0.1: icmp_seq=5 ttl=64 time=0.079 ms
64 bytes from 127.0.0.1: icmp_seq=6 ttl=64 time=0.041 ms
64 bytes from 127.0.0.1: icmp_seq=7 ttl=64 time=0.037 ms
64 bytes from 127.0.0.1: icmp_seq=8 ttl=64 time=0.031 ms
64 bytes from 127.0.0.1: icmp_seq=9 ttl=64 time=0.050 ms
64 bytes from 127.0.0.1: icmp_seq=10 ttl=64 time=0.047 ms
64 bytes from 127.0.0.1: icmp_seq=11 ttl=64 time=0.085 ms
64 bytes from 127.0.0.1: icmp_seq=12 ttl=64 time=0.051 ms
64 bytes from 127.0.0.1: icmp_seq=13 ttl=64 time=0.043 ms
64 bytes from 127.0.0.1: icmp_seq=14 ttl=64 time=0.045 ms
^C
--- 127.0.0.1 ping statistics ---
14 packets transmitted, 14 received, 0% packet loss, time 13320ms
rtt min/avg/max/mdev = 0.031/0.056/0.121/0.024 ms
```

ping <your default gateway> ·

```
venom@VENOM:~$ ping 10.230.76.191
PING 10.230.76.191 (10.230.76.191) 56(84) bytes of data:
64 bytes from 10.230.76.191: icmp_seq=1 ttl=64 time=4.36 ms
64 bytes from 10.230.76.191: icmp_seq=2 ttl=64 time=4.24 ms
64 bytes from 10.230.76.191: icmp_seq=3 ttl=64 time=8.61 ms
64 bytes from 10.230.76.191: icmp_seq=4 ttl=64 time=8.28 ms
64 bytes from 10.230.76.191: icmp_seq=5 ttl=64 time=171 ms
64 bytes from 10.230.76.191: icmp_seq=6 ttl=64 time=8.18 ms
64 bytes from 10.230.76.191: icmp_seq=7 ttl=64 time=11.8 ms
64 bytes from 10.230.76.191: icmp_seq=8 ttl=64 time=4.86 ms
64 bytes from 10.230.76.191: icmp_seq=9 ttl=64 time=4.40 ms
64 bytes from 10.230.76.191: icmp_seq=10 ttl=64 time=5.16 ms
64 bytes from 10.230.76.191: icmp_seq=11 ttl=64 time=5.82 ms
64 bytes from 10.230.76.191: icmp_seq=12 ttl=64 time=9.31 ms
64 bytes from 10.230.76.191: icmp_seq=13 ttl=64 time=8.89 ms
64 bytes from 10.230.76.191: icmp_seq=14 ttl=64 time=4.82 ms
64 bytes from 10.230.76.191: icmp_seq=15 ttl=64 time=11.4 ms
^C
--- 10.230.76.191 ping statistics ---
15 packets transmitted, 15 received, 0% packet loss, time 14019ms
rtt min/avg/max/mdev = 4.244/18.075/171.003/40.946 ms
```